

Dynamic evacuation path planning for subway station fire based on IACO

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ARTICLE INFO

Keywords:

Subway fire
Ant colony
FDS simulation
Evacuation path

ABSTRACT

Fire in subway stations is the most serious type of accident causing casualties, and traditional static evacuation paths are no longer able to guarantee evacuation safety. To solve this problem, this study proposes an **improved ant colony algorithm (IACO)**, which dynamically adjusts the evacuation paths by combining the simulation results of simulation software (PyroSim), and extends the available safe evacuation time of the personnel in a limited way. First, PyroSim is used to simulate the subway station hall and platform fire scenarios with the temperature, carbon monoxide (CO) concentration, and visibility at each location, obtained from twelve sets of monitors. Improvements in the pheromone concentration and update strategy of the ant colony algorithm (ACO) can improve the accuracy and convergence speed of the algorithm. Optimal path planning is achieved by the IACO, and then evacuation paths are dynamically adjusted by comparing the time at which each monitoring point on the path reaches the human hazard thresholds. Finally, until no adjustments can be made to obtain an evacuation path that extends the available safe evacuation time for personnel in a limited way. Compared to the traditional static evacuation path, this study provides a dynamic evacuation path that takes into account the real-time changes of fires in subway stations. This dynamic evacuation path planning approach will be put into use in subway stations before the accident to provide guidance on fire safety in subway stations and to provide contingency plans for emergency evacuation.

1. Introduction

With the development of the economy and the acceleration of the world's urbanization process, the existing transportation infrastructure is limited in the ground space that can accommodate the population, and the contradiction of urban traffic congestion becomes more and more prominent [1]. Due to the special underground operation mode, the input and use of rail transit greatly relieve the pressure of urban ground transportation and greatly facilitate people's travel [2]. At the same time, it has an important role in optimizing urban spatial structure and driving regional economic development [3]. The vigorous development of rail transportation also has an outstanding contribution to the promotion of ecological protection. However, subway stations are mostly built underground, with complex structures and relatively enclosed internal spaces, high passenger flow, and dense personnel. In the event of a fire, the entrances and exits of subway stations are limited, and the direction of personnel evacuation is consistent with the direction of smoke diffusion, making it difficult to detect the fire and evacuate personnel safely, which can easily lead to mass casualties and significant economic losses. Table 1 shows the subway fire accidents that have occurred in various countries around the world in recent

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years.

A subway station fire is a sudden and accidental accident, and the evacuation process should be simulated before the fire occurs to ensure that the evacuation path and strategy are adjusted [4]. In recent years, scholars in various fields have established fire models through FDS (Fire Dynamics Simulator) to analyze the impact of fire products in various buildings under fire conditions, and then simulate personnel evacuation and propose optimized evacuation plans or clarify the optimal evacuation planning [5–7]. TsukaharaM et al. [8] proposed a new evacuation path for subway station fires and found that a downward evacuation path opposite the direction of smoke diffusion is more conducive to safe evacuation. Li D C et al. [9] proposed opening gates and grid doors and other methods to reduce personnel evacuation time by studying the fire evacuation situation in subway stations. JinB et al. [10] analyzed the evacuation situation of subway pedestrians under multiple dangerous node failures and proposed three strategies that can achieve moderate evacuation time and crowd density. ZhangN et al. [11] used PyroSim and Pathfinder to find that the optimized spatial layout of the subway station can ensure the safe evacuation of personnel in the station under fire conditions. Biomimetic research, which simulates animal behavior and comprehensively considers the instinctive reactions and behaviors of evacuated objects, has been widely used in the field of personnel evacuation research. Nishinari et al. [12] modeled the positive feedback mechanism of ant foraging and used cellular automata simulation to elucidate the similarity between path selection and evacuation during ant foraging. Liu M et al. [13] proposed an improved quantum ant colony algorithm that can find evacuation plans faster with fewer individuals. Longzhen Z et al. [14] improved the ant colony algorithm convergence faster by combining artificial potential fields and quantum evolution theory. Xiao W et al. [15] proposed a fire escape planning method that effectively improves the algorithm's dynamic searchability and search speed by introducing the adaptively improving pheromone update strategy. Shuang S et al. [16] used a crowd density map to determine the location of crowded areas and used an improved ant colony algorithm to optimize the evacuation path with the shortest distance and the least turning point. Chuanming T et al. [17] established an emergency evacuation model with the Louvre as the research object, introduced the ant colony algorithm, and trained the BP neural network to identify potential evacuation bottlenecks and propose specific evacuation paths.

Currently, existing research focuses on fire simulation to analyze the influencing factors in the process of fire occurrence. The evacuation model solutions used are generally the shortest paths and do not consider the real-time effects of fire occurrence. Therefore, this study proposes a dynamic evacuation path planning method based on IACO, which takes into account the building of subway stations and changes in the fire environment, and dynamically adjusts the evacuation paths through real-time updated environmental information which can provide a theoretical basis and technical support for the safety management and plan selection of emergency evacuation subway stations.

2. Fire model

2.1. Theoretical analysis of subway fires

The safe evacuation of subway station fire personnel depends on the development of the fire and the evacuation time, as shown in Fig. 1. One is the Available Safety Egress Time (t_{ASET}), and the other is the Required Safety Egress Time (t_{RSET}). Only when $t_{RSET} < t_{ASET}$ is met can all personnel in the subway station be safely evacuated [18]. When t_{RSET} is constant, it becomes worth thinking about how to expand t_{ASET} .

t_{ASET} is determined by the rate of fire development and spread, as well as the flow and diffusion of smoke, and is the time from the occurrence of a fire to the critical point of human danger in the environment inside the subway station. Fire environments that exceed the critical point of danger can cause serious harm to human health. The endurance limit of personnel in a fire environment usually considers factors such as temperature, visibility, and CO concentration. t_{RSET} refers to the time from the occurrence of a fire to the complete safe evacuation of personnel, including the detection and reporting of fire alarm time (T_{alarm}), personnel pre-action time (T_{pre}), and completion of personnel emergency evacuation time (T_{move}). T_{move} refers to the evacuation time required for personnel to evacuate to a safe location, which largely determines the duration of t_{RSET} [19].

2.2. Fire scene layout

A detailed analysis of the subway, a special underground building, reveals that its most significant characteristics are high personnel density and traffic flow. Once a fire or other disaster occurs in a subway station, compared to the same accident occurring in other buildings, its condition is more difficult to control and the consequences will be more serious. Therefore, this article selects subway stations as the research object and establishes an intelligent planning method for the emergency evacuation path of subway

Table 1
Subway fire accidents in recent years in various countries around the world.

Time	Site	Accident
June 2013	Moscow Metro, Russia	A fire in the power cord caused a fire that injured 66 people
January 2015	Washington D.C. Metro, USA	A short circuit in the power supply line caused a fire, killing one person and injuring 84 others
January 2016	Tokyo Subway, Japan	The unknown substance burned without causing casualties
February 2017	Hong Kong MTR	The unknown substance burned without causing casualties
March 2020	New York subway, USA	Deliberately set fire, 18 people were injured
January 2021	Mexico Metro	Equipment failure caused a fire, killing one person and injuring 30 others
June 2021	London Underground, UK	A fire broke out, injuring 6 people
November 2021	Tokyo Subway, Japan	Deliberate arson and striking resulted in 17 injuries

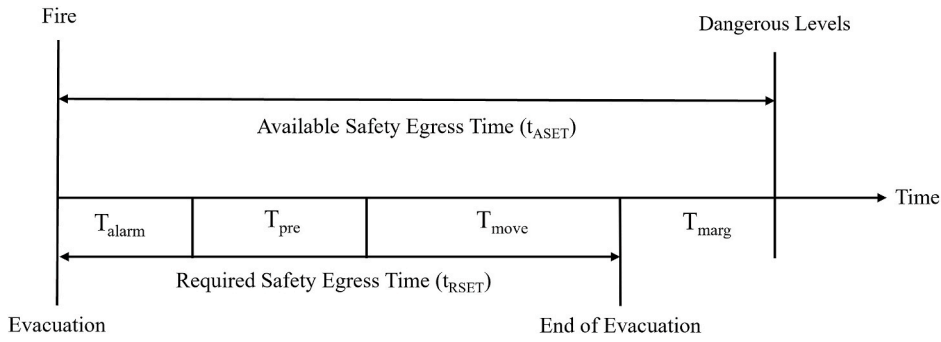


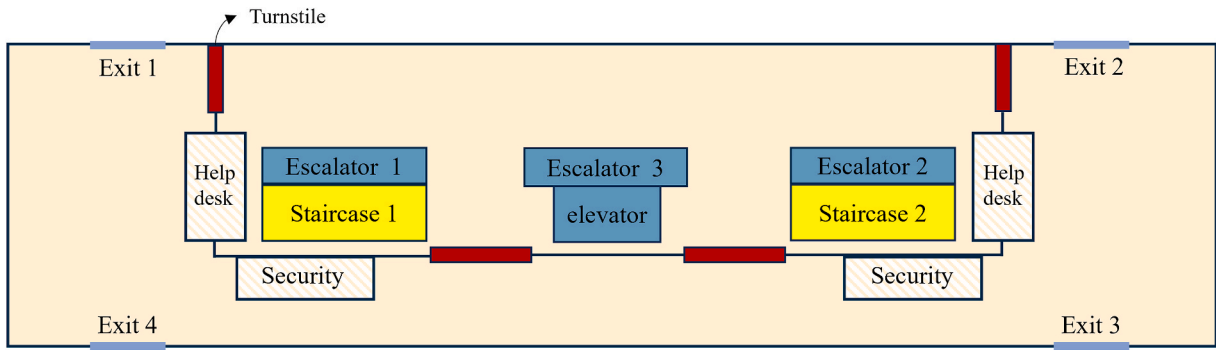
Fig. 1. Relationship between fire development and evacuation time.

station fire personnel.

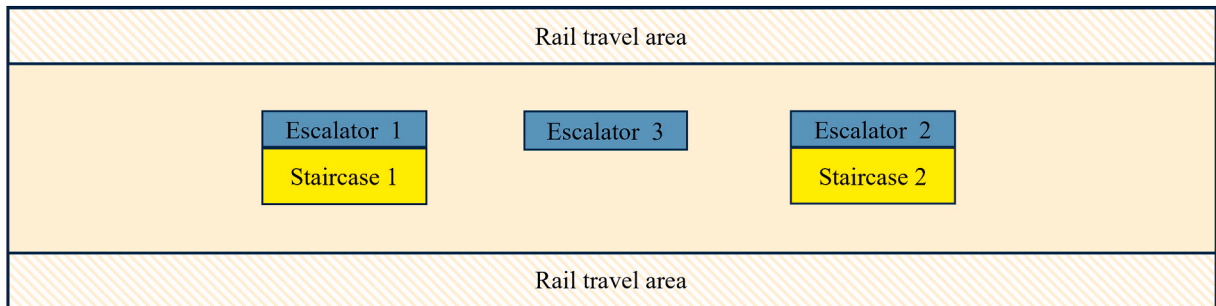
The architectural structure of the simulated subway station model is shown in Fig. 2. The first underground floor is the station hall, and the second underground floor is the platform. The platform floor is connected to the station hall floor by three escalators, one vertical ladder, and a step ladder on both sides; The station hall floor is connected to the outside and has four entrances and exits. There are a certain number of gates, manual windows, security inspection instruments, pillars, and other infrastructure in the paid area.

PyroSim, a great graphic user interface developed by Thunderhead Engineering, was used to help complete fire simulation simulation experiments in subway stations in this study. The core of the software is FDS, which can be established for real fire simulations in various building environments and to calculate smoke diffusion, temperature changes, and toxic gas concentration changes in complex environments [20,21].

Based on comprehensive research needs and software requirements, the paid areas of subway stations where passengers may evacuate are divided into a grid of 0.5 m * 0.5 m * 0.5 m, and fire sources are set up at the station hall and platform floors. There are many burning materials in subway stations, and this study is designed to cause fires due to the combustion of flammable items in passenger luggage. According to Hansell, G. O. et al. the heat release rate from fires in crowded places such as stations is in the range of 2.0–3.5 MW [22]. The heat release rate of the fire source is set to 3 MW. The fast fire is selected for calculation and simulated with the t^2 fire development model, with the corresponding fire growth coefficient of 0.1878 kW/s² the heat release rate of the fire tends to stabilize when the time is 126.39s. According to GB50157-2003, the evacuation time for personnel is 6 min, and the simulation time is



(a) Subway station hall floor



(b) Subway station platform floor

Fig. 2. Building structure diagram of metro station.

set to 360s [23].

$$q = at^2 \quad (1)$$

Where q is the heat release rate of the ignition source (kW); a is the fire development coefficient (kW/s²); t is the development time of the fire (s).

The location of fires and the distribution of fire product monitoring points in the station hall and platform floor of this study are shown in Fig. 3. The traditional bottom-up conventional evacuation method was used in this study. The evacuation direction is considered to be the upward evacuation of passengers from the platform floor to the station hall floor through the steps, and then passengers reaching the station hall floor are evacuated to the exits through the gates, which determines the selection of the monitor locations in the fire simulation model. A group of temperature, CO concentration, and visibility detection devices are placed at four entrances and exits, two sets of step exits on the station hall floor, and two sets of step exits on the platform floor, totaling twelve sets.

This paper uses the formula proposed by Milke [24] to calculate the human hazard threshold, which is related to the flue gas temperature impact factor $f_1(T)$, the visibility impact factor $f_2(\rho_{CO})$ and the carbon monoxide impact factor $f_3(K_c)$.

$$f_1(T) = \begin{cases} 1 & T_0 < T_s \leq T_{c1} \\ (v_{max} - 1.2) \left[\frac{T_s - T_{c1}}{T_{c2} - T_{c1}} \right]^2 / v_0 + 1 & T_{c1} < T_s \leq T_{c2} \\ v_{max}^{1.2} \left[1 - \left[\frac{T_s - T_{c1}}{T_{c2} - T_{c1}} \right]^2 \right] & T_{c2} < T_s \leq T_d \end{cases} \quad (2)$$

$$f_2(\rho_{CO}) = \begin{cases} 1 & \rho_{CO} < 0.1 \\ 1 - (0.2125 + 1.788\rho_{CO}t) & 0.1 \leq \rho_{CO} < 0.25 \\ 0 & \rho_{CO} \geq 0.25 \end{cases} \quad (3)$$

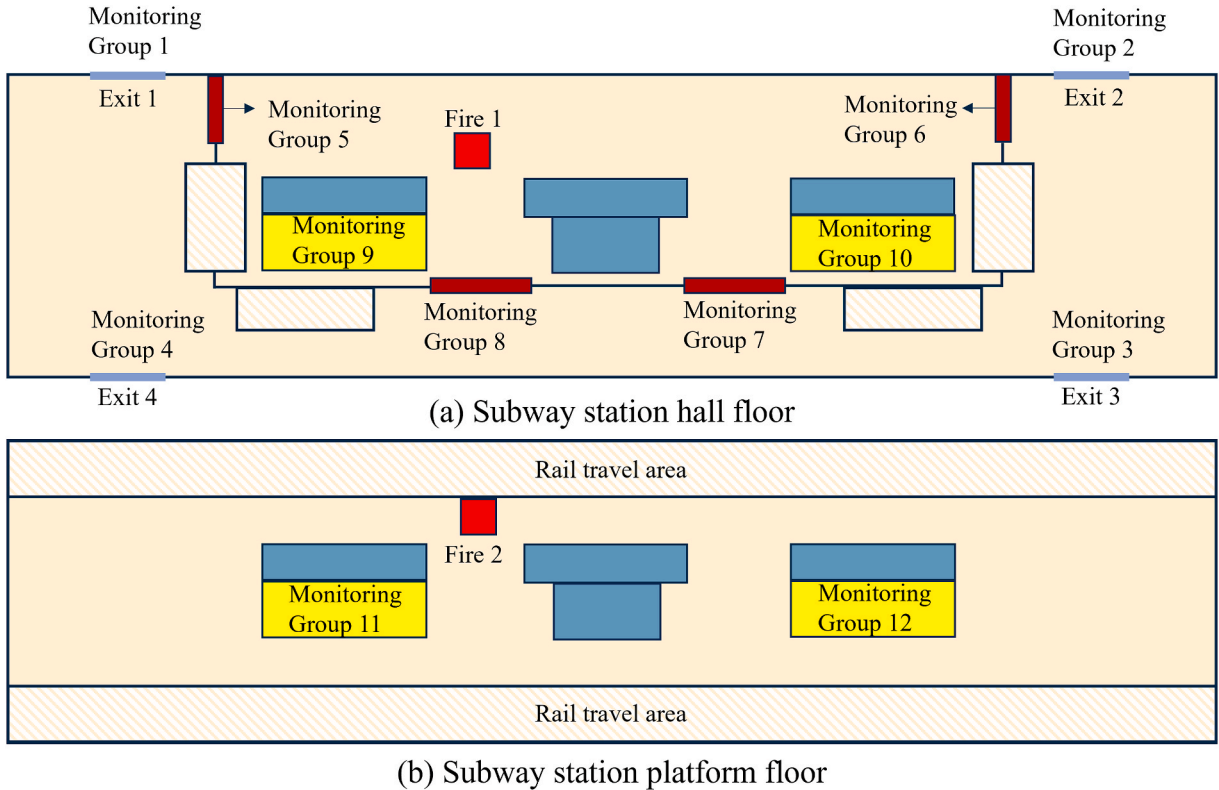


Fig. 3. Fire scenario and monitoring point layout.

$$f_3(K_c) = \begin{cases} 1 & K_c < 0.35 \\ 1 - 0.3025K_c & 0.35 \leq K_c < 1.2 \\ 0.25 & K_c \geq 1.2 \end{cases} \quad (4)$$

where T_s is the temperature at the fire scene ($^{\circ}\text{C}$); T_0 is the room temperature (20°C); v_0 is the normal walking speed (set as 1.2 m/s); $v_{\max}$ is the maximum walking speed (set as 5 m/s); T_{c1} is the temperature causing discomfort (taken as 30°C); T_{c2} is the temperature causing an injury (taken as 60°C); and T_d is the lethal temperature (set as 120°C). When the temperature exceeds 120°C , $f_1(T)$ is 0; ρ_{CO} is the concentration of carbon monoxide, t is the time of exposure to carbon monoxide, $f_2(\rho_{\text{CO}}) = 0$ indicates that a person faints or dies; K_c is the extinction coefficient ($1/\text{m}$), and the extinction coefficient, which is positively related to the concentration of the smoke, and inversely related to the visibility.

Milke smoke temperature impact coefficient formula shows that when the temperature is less than 30°C , there is no effect on the speed of people moving; when the temperature is greater than 30°C less than 60°C , the personnel begin to feel uncomfortable, but the temperature did not cause harm to the human body, at this time the higher the temperature, the faster the speed of people moving; when the temperature is greater than 60°C began to cause harm to the human body, the speed of people moving with the increase in temperature is reduced, once the temperature reaches 120°C , the speed of the personnel moving to 0 m/s .

when the concentration of carbon monoxide is greater than 0.1% or less than 0.25% , the longer the personnel and the hazardous gases are exposed to the longer the accumulation of hazardous and toxic substances in the body, the greater the harm caused to the human body. The more harm will be caused to the human body. Therefore, the concentration of carbon monoxide is higher than 0.1% as the critical value of the harm caused to human beings.

The results of many existing studies show that the light reduction coefficient in a small space fire cannot be lower than 0.2 and visibility cannot be lower than 5 m ; the light reduction coefficient in a large space fire cannot be lower than 0.1 and visibility cannot be lower than 10 m . In case of a fire in a subway station, a visibility of less than 10 m is considered as the human hazard threshold.

This study determined that the critical value of harm to human health caused by fire critical influences is:

- 1) The temperature of air and flue gas below 1.6 m shall not exceed 120°C
- 2) The CO concentration in air below 1.6 m shall not exceed 0.1%
- 3) The visibility with an average eye height of 1.6 m shall not be less than 10 m .

2.3. Fire on the subway station Hall

The simulation of smoke diffusion in the station hall floor fire is shown in Fig. 4. Due to the rapid upward diffusion of smoke plumes caused by the fire, they touched the top of the subway station hall, forming a top-level jet effect and starting to spread laterally. As the fire developed, smoke gradually settled downwards to form thick smoke, but the impact of thick smoke on the platform floor was limited.

In the event of a station floor fire, the changes in 2D slice data for the fire critical influences of the fire at a height of 1.6 m from the station floor are shown in Fig. 5:

- 1) The temperature of the station hall floor continuously increased with the time of the fire, but it did not exceed 120s except for the fire source point. At the same time, the temperature of the station hall floor was slightly higher on both sides than in the middle, which was caused by the rapid diffusion and settlement of smoke towards both sides.
- 2) The CO concentration in the station hall gradually increased over time but did not reach the dangerous critical value within the simulation time.
- 3) Due to the diffusion of smoke on both sides, the visibility on both sides of the station hall floor decreased at the first moment. When the fire occurred for 120s , the visibility on the station hall floor had generally decreased to below the dangerous critical value of 10 m .

Simulation results are shown in Fig. 6. Fig. 6(a) shows the most variable temperature changes at monitoring points 2 and 3, which reached 92°C and 93°C respectively at 360s . However, the impact of a fire in the station hall floor on the temperature of the platform

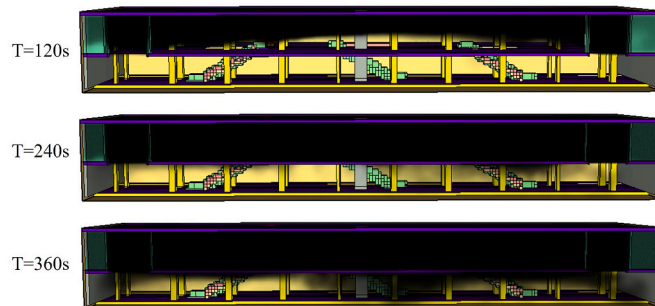


Fig. 4. Smoke diffusion situation in the station hall floor fire.

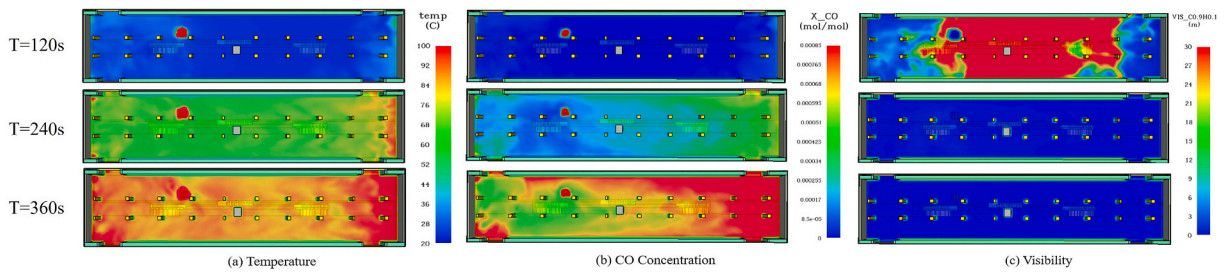


Fig. 5. Changes in fire critical influences at different times in the station hall floor fire.

floor was not significant. Within 360s of fire simulation, the temperature of each monitoring point did not reach the dangerous critical value of 120 °C. Fig. 6(b) shows that the largest change in CO concentration is at monitoring point 2, which reached the dangerous critical value of 0.1% at 350s. Fig. 6(c) shows that the visibility changes at each monitoring point. The fire on the station hall floor has a significant impact on the monitoring point 10 distributed on the station hall floor, with visibility reduced to below the dangerous critical value. The visibility of monitoring points 11 and 12 remained unchanged during the simulation time.

The time when each monitoring point on the station hall floor reaches the critical danger value is listed in Table 2. The most significant impact of fire critical influences on the station hall floor is the change in visibility, with multiple monitoring points reaching dangerous critical values. When a fire occurs on the station hall floor, the smoke diffuses upwards and has relatively little impact on the temperature, CO concentration, and visibility of the platform floor on the underground second floor.

2.4. Fire on the subway station platform

The simulation of smoke diffusion in the station platform floor fire is shown in Fig. 7. Due to the formation of the top layer jet effect, the thick smoke caused by the platform floor fire spreads laterally and also spreads towards the station hall through the opening of the escalator and step ladder. With the development of the fire, thick smoke gradually occupied the space on the two floors of the station hall and platform.

In the event of a station platform fire, the changes in 2D slice data for the fire critical influences of the fire at a height of 1.6 m from the station platform are shown in Fig. 8:

- 1) The temperature of the platform floor exceeded the critical value of danger. Affected by the location of the fire, the temperature of the 360s platform floor shows that the left side was higher than that on the right side.
- 2) The CO concentration on the platform floor did not reach the dangerous critical value.
- 3) Due to the first diffusion of smoke to both sides, the visibility on both sides of the station hall floor decreased at the first moment. With the settling of smoke, the visibility on the platform floor quickly decreased to below 10 m.

Simulation results of monitoring points in the station platform floor fire are shown in Fig. 9. Fig. 9(a) shows the temperature change at monitoring point 11 reached above 65 °C during the fire simulation time, while the temperature at other monitoring points was between 50 °C and 60 °C. The strongest changes in CO concentrations were observed at monitoring 11 and 12, as shown in Fig. 9(b). Fig. 9(c) shows the thick smoke quickly spread to the station hall and platform floor, causing the visibility of monitoring points to gradually decrease to below the dangerous critical value of 10 m. The visibility of monitoring point 4 located on the station hall floor has been reduced to 10 m at 97s.

The time when each monitoring point of the platform floor fire reaches the critical danger value is listed in Table 3. Under the same fire conditions, the temperature and CO concentration changes caused by platform floor fire are relatively small compared to station

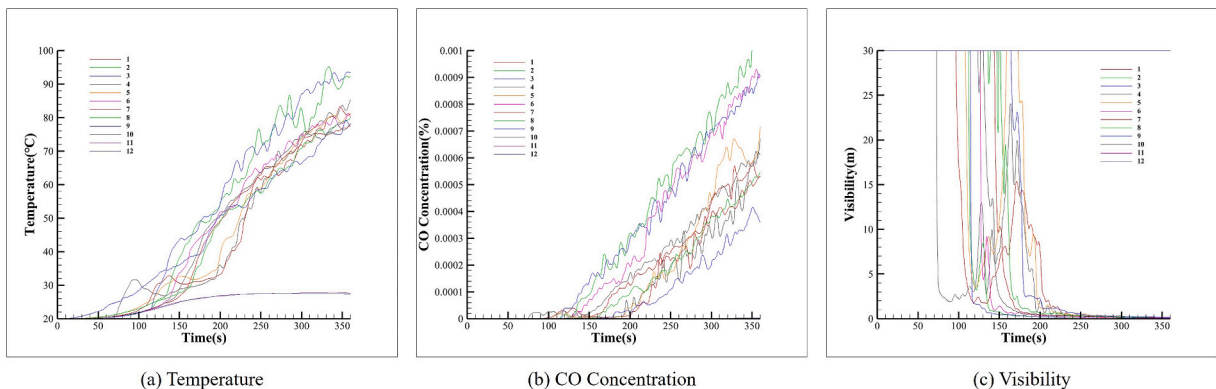


Fig. 6. Simulation results of monitoring points in the station hall floor fire.

Table 2
The critical time of danger at each monitoring point of fire in the station hall floor.

Monitoring points	1	2	3	4	5	6	7	8	9	10	11	12
Temperature												
CO Concentration		350s										
Visibility	104s	114s	115s	285s	74s	112s	128s	148s	162s	178s		

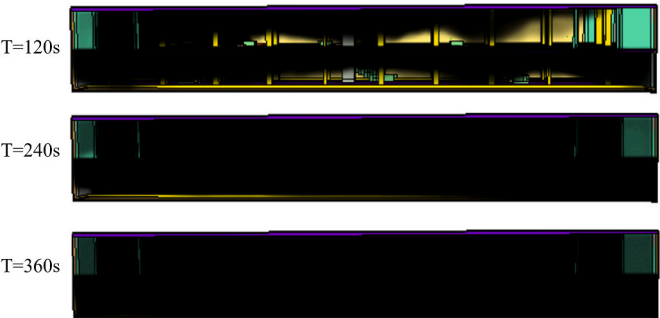


Fig. 7. Smoke diffusion situation in the platform floor fire.

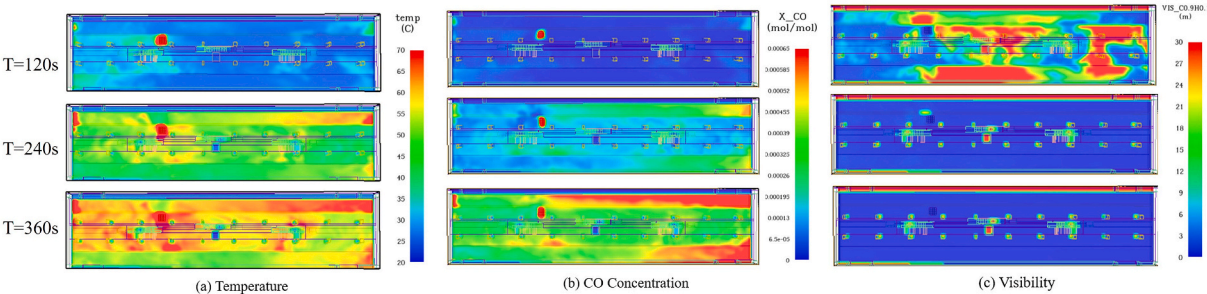


Fig. 8. Changes in fire critical influences at different times in the station platform floor fire.

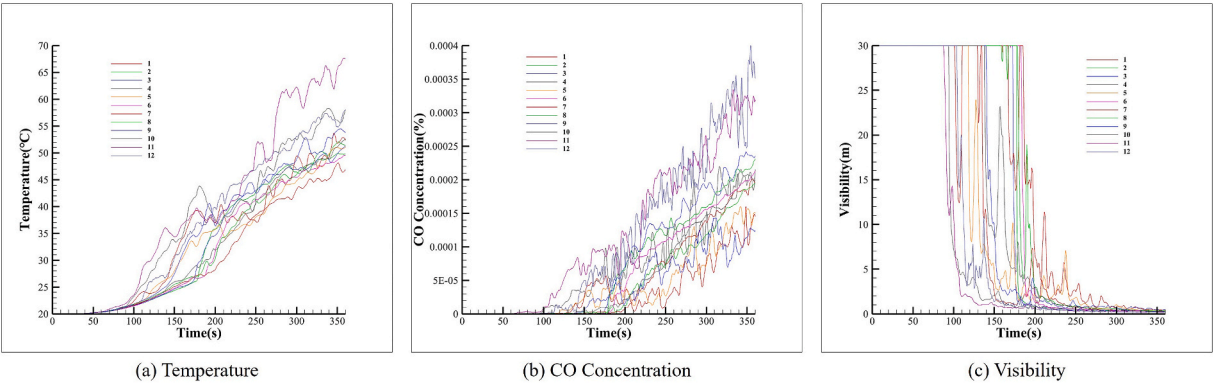


Fig. 9. Simulation results of monitoring points in the station platform floor fire.

Table 3
Critical time of danger at each monitoring point of platform floor fire.

Monitoring points	1	2	3	4	5	6	7	8	9	10	11	12
Temperature												
CO Concentration												
Visibility	140s	178s	175s	97s	120s	186s	198s	182s	150s	143s	100s	113s

hall floor fire. However, the change in visibility is the most different. This is because platform floor fire will diffuse upwards through escalators and step entrances to the station hall floor, resulting in more smoke accumulation at the station hall floor, and the smoke on the station hall floor will decrease faster.

3. Evacuation model

Evacuation methods that are more widely used, such as Helbing's social force model, cellular automata (CA) model, and agent model, are limited by individual factors and are unable to comprehensively and dynamically consider group evacuation situations. To better explore the path selection of evacuation groups in practical situations, this study improves the ant colony algorithm for path optimization.

3.1. Algorithm comparison

Pathfinder simulation software is an agent-based simulation software for personnel evacuation research, which uses the A* search algorithm and combines string-pulling technology [25]. The passable area of the Pathfinder model is divided into three-dimensional triangular mesh areas as shown in Fig. 10(a), which can accurately represent geometric details and curves, allowing evacuation paths at any angle to directly evacuate towards the target point. The ACO path is composed of line segments parallel to or at an angle of 45° to the coordinate axis, as shown in Fig. 10(b), and can only disperse in a direction parallel to or at an angle of 45° to the coordinate axis. This results in ACO path planning being longer than the Pathfinder software. Therefore, without considering the dynamic development of fires, the reliability of the ACO is compared and verified using the Pathfinder software path planning results as a reference.

A grid format map is established with three exits of 20×20 , with red dots indicating the starting point of evacuation personnel, green markings indicating the three exit positions, and black indicating obstacles as shown in Fig. 11. Select the same evacuation starting point, conduct evacuation simulation using Pathfinder software, and complete path planning using the ACO developed in Matlab. The comparison of evacuation results is shown in Fig. 12(a) and (b).

By comparison, it was found that the average length of the simulated evacuation path for multiple groups of personnel in Pathfinder only chose the nearest exit, with paths largely overlapping each other and not fully considering the other two exits, which can easily cause congestion and increase the casualty rate during the actual evacuation process. On the contrary, the result of ACO fully considers three exits in path planning, but the path planned by this algorithm is relatively long, and the convergence speed is too slow. Based on the above analysis, improvements are made to address the shortcomings of ACO being trapped in local optima and slow iteration speed.

3.2. Establishment of the algorithm model

3.2.1. Principle of the algorithm

Marco Dorigo proposed ACO by observing the foraging behavior of ants in nature. ACO searches for the optimal solution through a heuristic exploration strategy and positive feedback generated by the collaboration between ant colony individuals [26]. Because ACO uses information exchange between ants for path planning, similar to the information exchange mechanism between evacuees, ACO is chosen for research.

1) Parameter introduction

The performance of ACO is directly influenced by parameters. α reflects the degree to which the concentration of pheromones accumulated on the path of ants during their foraging and pathfinding process affects their path selection. β indicates the degree to which heuristic information affects ants' path selection. ρ indicates that the pheromone of ants will gradually decrease over time. Q indicates the total amount of pheromones released by ants on the path after completing one traversal. m indicates the total amount of ant population, which will affect the stability and global search ability of algorithms.

2) Heuristic function

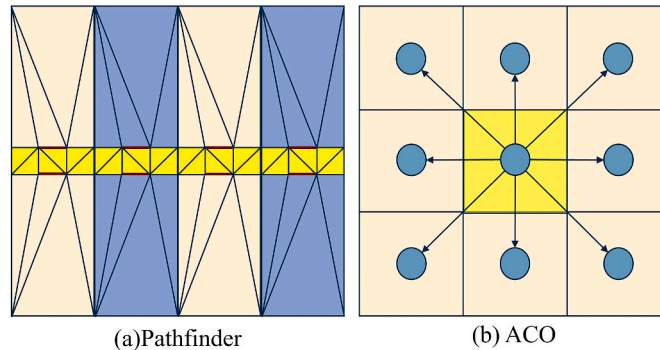


Fig. 10. Comparison of Pathfinder and ACO algorithm planning path node selection methods.

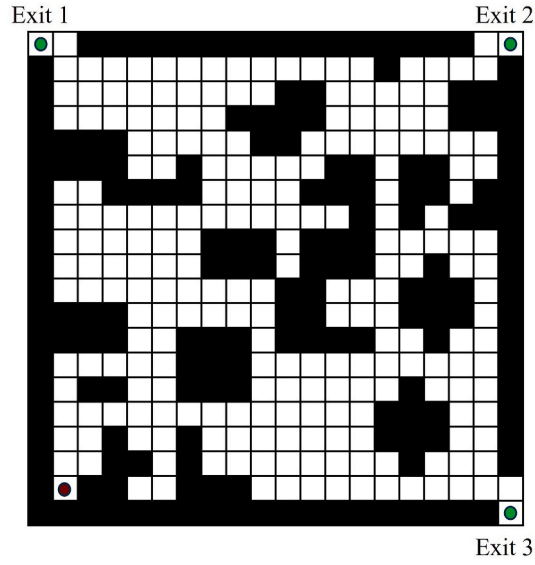


Fig. 11. Schematic diagram of a 20 * 20 small-sized model.

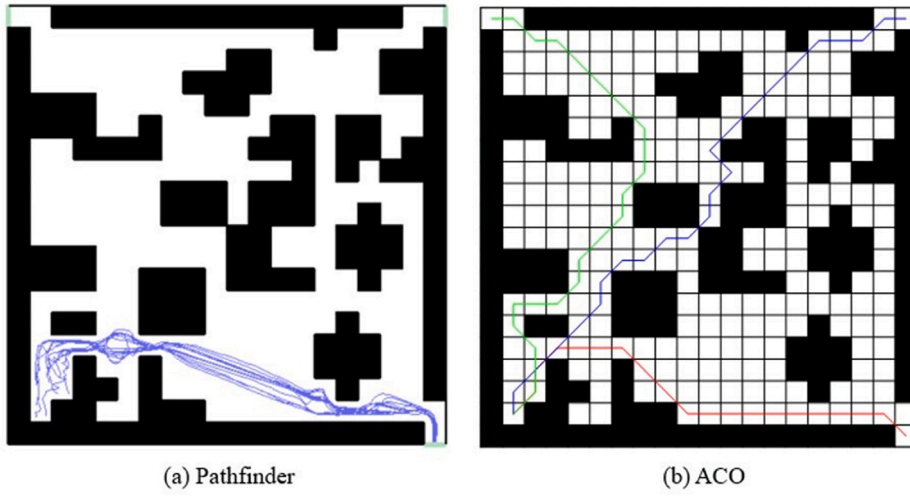


Fig. 12. Comparison of pathfinder and ACO algorithm path planning.

The heuristic function in the basic ACO indicates that a smaller distance between i and j results in a greater probability of choosing this path.

$$\eta_{ij}(t) = 1/d_{ij}(t) \quad (5)$$

3) Transfer probability

At time t , the ant k at i will use a roulette selection strategy to select j , and the probability of selecting the node is represented by $P_{ij}^k(t)$:

$$P_{ij}^k(t) = \begin{cases} \frac{\tau_{ij}^\alpha(t)\eta_{ij}^\beta(t)}{\sum_{s \in allowed_k} \tau_{is}^\alpha(t)\eta_{is}^\beta(t)} & , j \in allowed_k \\ 0 & , j \in unallowed_k \end{cases} \quad (6)$$

Where $\tau_{ij}(t)$ represents the pheromone concentration on the path from i to j . $allowed_k$ represents a set of nodes that ants k ($k = 1, 2, \dots, m$)

have not traversed. s is a node belonging to that set.

4) Pheromone update

If the $\tau_{ij}(t)$ is too large, it will cover the impact of $\eta_{ij}(t)$ on ant path selection. Therefore, the pheromone quantity on the path needs to be updated after each iteration.

$$\tau_{ij}(t+1) = (1-\rho)\tau_{ij}(t) + \Delta\tau_{ij} \quad (7)$$

$$\Delta\tau_{ij} = \sum_{k=1}^m \Delta\tau_{ij}^k \quad (8)$$

Where $\Delta\tau_{ij}^k$ is the pheromone released by ant k from i to j . $\Delta\tau_{ij}$ is the sum of pheromones released by all ants from i to j .

5) Update policy

The ACO model includes the ant cycle model, the ant quantity cycle model, and the ant density cycle model. This study uses an ant cycle model that adopts a global update strategy and is suitable for path planning. The formula of this model is as follows:

$$\Delta\tau_{ij}^k = Q/L_k \quad (9)$$

Where L_k is the total length of the path traveled by ant k .

3.2.2. Improvement of the algorithm

Firstly, the pheromone update strategy is improved through adaptive adjustment of the ρ value to balance the global search performance and convergence speed of the algorithm. When the optimal solution of the algorithm has not significantly improved after n iterations, reduce the ρ value to improve the algorithm's global search ability, and set the ρ_{Min} value to avoid reducing convergence speed.

$$\rho(t) = \text{Max}\{0.95\rho(t-1), \rho_{Min}\} \quad (10)$$

The intensity of pheromones is a constant that affects algorithm stagnation. During the calculation process, if the optimal solution of the algorithm does not change for a while, the search results of the algorithm may fall into a certain extreme value [27]. However, this extreme value may not necessarily be the global optimal solution. Therefore, this study refers to the improvement of the pheromone update strategy based on the ant cycle model to replace the constant total amount of pheromones Q with a time function $Q(t)$ to achieve dynamic adjustment of pheromone intensity and avoid the algorithm falling into local optima [28].

$$Q(t) = \begin{cases} Q & K < K_{Max}/5 \\ Q/3 & K_{Max}/5 \leq K < 3K_{Max}/5 \\ Q/5 & 3K_{Max}/5 \leq K < K_{Max} \end{cases} \quad (11)$$

Where K is the current number of iterations and K_{Max} is the maximum number of iterations.

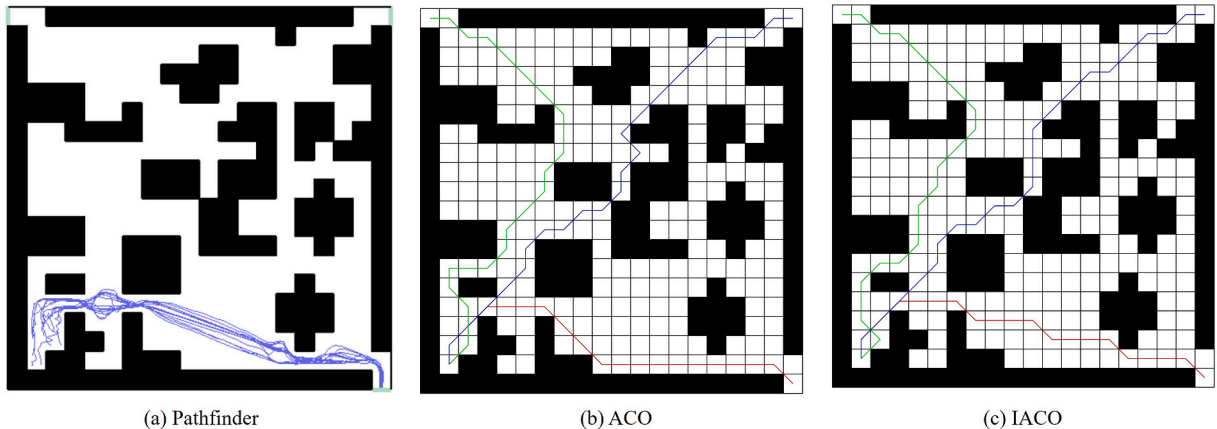


Fig. 13. Comparison of Pathfinder, ACO, and IACO algorithm path planning.

Using IACO to complete path planning on the same 20 m* 20 m grid format map, the resulting path is compared with the two sets of methods mentioned above, as shown in Fig. 13. The comparison of the distance and algorithm iteration times required for ACO and IACO to evacuate to different exits is shown in Table 4.

It can be seen that the distance of the IACO evacuation path from Exit 3 is 21.48 m, which is close to the average distance of the Pathfinder software evacuation results. Compared to ACO, IACO not only shortens the distance of evacuation path planning but also significantly reduces the number of iterations for algorithm convergence which effectively improves the positive feedback effect and convergence speed of the basic ant colony algorithm, avoiding algorithm stagnation and falling into local optimal solutions.

4. Algorithm simulation experiment

Based on IACO, a personnel evacuation simulation platform is established by Matlab. The simulation evacuation layout of the station hall and platform floors is shown in Fig. 14. Determine the shortest path for personnel evacuation, and then analyze the fire critical influences obtained from the PyroSim fire simulation to dynamically adjust the evacuation path and propose personnel evacuation plans under different fire conditions.

4.1. IACO parameter analysis experiment

The determination of the main parameter values of the algorithm is crucial, and it will have varying degrees of impact on algorithm performance. Determine a set of parameter values ($\alpha = 1, \beta = 5, \rho = 0.5, Q = 1, m = 50$). Each time only one parameter value is changed to control the variable, and each group of parameter tests is conducted 10 times to obtain the optimal parameter combination for IACO in the evacuation environment.

1) Combination of α and β

Taking $\alpha \in \{0.5, 1, 1.5, 2, 5\}$ and $\beta \in \{5, 8, 10, 15, 20\}$, each parameter combination simulated 10 times to obtain the average are listed in Table 5. When $\alpha \approx 1$, the algorithm plans the shortest evacuation distance. When $\beta \geq 15$, the algorithm convergences. Further simulation results of different combinations of α and β are listed in Table 6. Then, taking $\alpha = 1, \beta = 12$ is the optimal combination.

2) Combination of ρ, Q , and M

Taking $\rho = \{0.4, 0.5, 0.6\}$, $Q = \{1, 50, 100, 150\}$ and $m = \{50, 100, 150\}$, different combinations of ρ, Q , and M are used for simulation, each parameter combination is simulated 10 times to get the average, and a total of 360 simulations are performed. The simulation results are given in Table 7.

4.2. Evacuation path adjustment

This study adopts the traditional evacuation path, which is the platform floor to station hall floor exit path for evacuation. In the event of a fire in a subway station, passengers on the platform floor will be evacuated from points A and B in the middle of the platform to points C and D in the station hall via the escalators (elevators are not available at this time). When a fire occurs, due to the layout structure of the subway station and passenger travel habits, many passengers are concentrated in the middle of the platform floor. The layout of ignition points and evacuation nodes for the subway station model is shown in Fig. 15.

4.2.1. Station hall fire

When a fire occurs on the station hall floor, passengers on the platform floor evacuate by walking up the stairs on both sides to the station hall floor. The evacuation path for fire personnel on the station hall floor is shown in Fig. 16, and the specific path length is shown in Table 8.

Path ① passes through monitoring points 1, 5, 9, and 11. Monitoring point 5 has the shortest time to reach the dangerous critical value, which is 94s. Therefore, the turnstile of monitoring point 5 cannot be used after 94s of evacuation, and the evacuation path will be dynamically adjusted. The evacuation path on the platform floor remains unchanged, and the adjustment results of the evacuation path on the station hall floor are shown in Fig. 17.

The adjusted optimal evacuation path is A—Staircase 1—C—Exit 4, with a path length of 59.456 m. At this point, the evacuation path passes through monitoring points 4, 8, 9, and 11. The shortest time for monitoring point 8 to reach the critical value of danger is 148s. The data from multiple monitoring points 148s after the fire occurred indicates that there is no feasible evacuation path on the station hall floor.

Path ② passes through monitoring points 2, 6, 10, and 12. Monitoring point 6 has the shortest time to reach the dangerous critical value, which is 112s. After 112s of evacuation, the evacuation path will be dynamically adjusted as B—Staircase 2—D—Exit 3, with a path length of 59.456 m, as shown in Fig. 18.

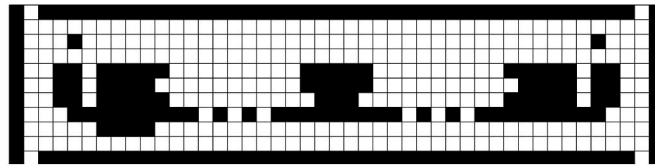
The adjusted optimal evacuation path passes through monitoring points 3, 7, 10, and 12. Monitoring point 3 reached a dangerous critical value at 115s. The path changes as B—Staircase 2—D—Exit 4, with a path length of 97.1128 m, as shown in Fig. 19.

The adjusted optimal evacuation path passes through monitoring points 4, 7, 10, and 12. The turnstile of monitoring point 7 failed 128s after the fire occurred. Adjust the evacuation path again, as shown in Fig. 20. This path passes through monitoring point 8 with a critical time of 148s, so there is no feasible evacuation path after 148s of the fire.

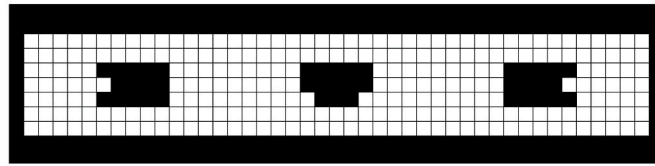
Therefore, when a fire occurs on the station hall floor, the available time for evacuations is 148s. Compared to the value of t_{ASET} obtained from fire simulation, the result of using the IACO has a longer available time for personnel evacuation, which dynamically

Table 4
Comparison of ACO and IACO planning path distance and iteration times.

Exit	Exit 1		Exit 2		Exit 3	
	Distance(m)	Number of iterations	Distance(m)	Number of iterations	Distance(m)	Number of iterations
ACO	35.38	162	29.21	47	23.72	45
IACO	25.97	10	28.38	11	21.48	7



(a) Subway station hall floor layout



(b) Subway station platform floor layout

Fig. 14. Simulated evacuation layout of station hall and platform floors.

Table 5
 α , β Parameter simulation based on statistics.

α	0.5	1	1.5	2	5
Average length	88.6205	83.7519	87.8205	93.0774	97.3200
B	5	8	10	15	20
Average length	83.7519	79.4122	77.1897	73.4558	73.4558

Table 6
Different α and β simulation results of parameter combinations.

parameter combinations	Average length					
	$\beta = 10$	$\beta = 11$	$\beta = 12$	$\beta = 13$	$\beta = 14$	$\beta = 15$
$\alpha = 0.5$	77.9813	77.2700	74.2073	72.7162	70.4736	70.3079
$\alpha = 1.0$	77.1897	76.4583	73.4558	73.4558	73.4558	73.4558
$\alpha = 1.5$	79.4926	78.3210	78.3009	77.7352	76.1754	75.0156

Table 7
Different ρ , Q , and m simulation results of parameter combinations.

parameter combinations		Average length			
ρ	Q	$m = 50$	$m = 100$	$m = 150$	
0.4	1	74.4014	72.2960	72.6274	
	50	70.9706	70.6392	71.1363	
	100	72.1304	74.8499	73.7872	
	150	74.9470	73.2901	74.1185	
0.5	1	73.4558	73.1931	71.9647	
	50	70.3079	70.3079	70.3079	
	100	70.4736	72.4617	73.0759	
	150	72.6274	73.9043	73.7872	
0.6	1	76.1270	73.9529	72.7931	
	50	70.6392	70.1422	70.4736	
	100	71.4677	71.6333	72.7931	
	150	72.79308	72.95876	73.6215	

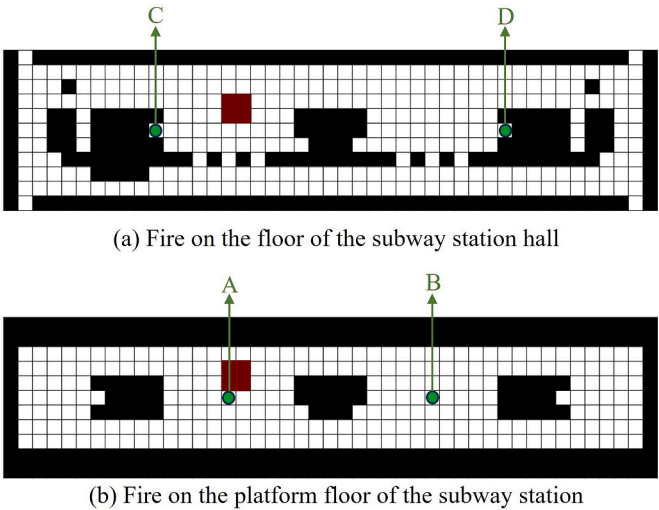


Fig. 15. Layout of ignition points and evacuation nodes of the subway station model.

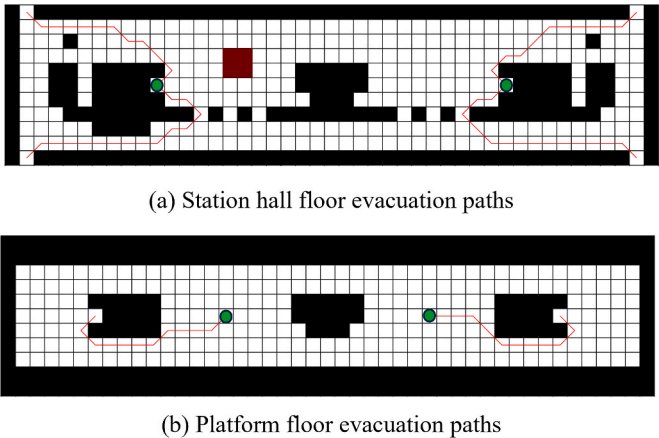


Fig. 16. Optimal evacuation path for fire personnel on the Hall floor.

Table 8
Optimal evacuation path length for station hall fire.

Path	Evacuation path	Distance(m)
①	A—Staircase 1—C—Exit 1	51.456
②	B—Staircase 2—D—Exit 2	51.456

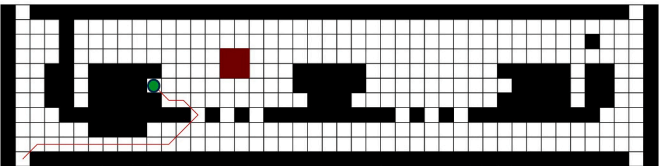


Fig. 17. Adjustment results of evacuation path on the station hall floor after 94s of fire.

adjusts and clarifies the optimal evacuation path.

4.2.2. Station platform fire

When a fire occurs on the platform floor, the evacuation path for personnel is shown in Fig. 21, and the specific path length is shown

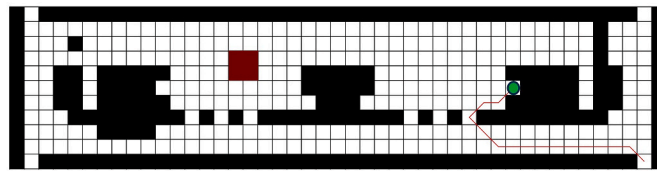


Fig. 18. Adjustment results of evacuation path on the station hall floor after 112s of fire.

in Table 9. Path ③ passes through monitoring points 1, 5, 9, and 11. Monitoring point 11 has the shortest time to reach the dangerous critical value, which is 100s.

Therefore, after a fire on the platform floor occurs for 100s, it is not possible to evacuate by walking up Staircase 1, and passengers on the platform floor are all passing through Staircase 2. The path changes as A—Staircase 2—D—Exit 2, with a path length of 82.7679 m, as shown in Fig. 22. Staircase 2 cannot pass after 113s, so the available evacuation time for this path is 113s. Path ④ passes through monitoring points 2, 6, 10, and 12. Monitoring point 12 has the shortest time to reach the dangerous critical value, which is 113s. Equally, Staircase 2 cannot pass after 113s, so the available evacuation time for this path is 113s.

When a fire occurs on the platform floor, considering the evacuation sequence from the platform floor to the station hall floor to the exit, it is particularly important to ensure that the passage time of the stairs on the platform floor is sufficient. Previously, the value of t_{ASET} obtained from fire simulation was 97s. The available time for personnel evacuation can be increased to 113s by using IACO for this fire condition.

5. Conclusion and outlook

In this study, fire simulation on the hall floor and platform floor of a subway station was performed to obtain the temperature, CO concentration, and visibility changes at each monitoring point during the fire. The pheromone concentration and update strategy of ACO are improved and the parameter configuration is completed for IACO. The evacuation path planned by IACO is dynamically adjusted according to the time of reaching the human danger threshold at each monitoring point. Eventually, an optimal evacuation path is obtained by this research method, which can take into account the real-time changes in the fire environment and effectively extend the available safe evacuation time of the personnel.

- 1) Based on PyroSim, fire simulations were conducted for different fire scenarios in the subway station hall and platform floors, clarifying the smoke diffusion characteristics and fire product variation patterns under each fire scenario, and determining the main factors affecting the available safe evacuation time of personnel under each fire scenario.
- 2) Clarify the basic principles of the ACO and improve it to avoid premature convergence and falling into local optima. Based on the IACO, a subway station fire evacuation model is constructed, and the parameter configuration is determined. The IACO is used to obtain the optimal evacuation path of personnel at each fire danger point under different fire scenarios.
- 3) The evacuation path is dynamically adjusted by comparing the time to reach the human hazard threshold at the monitoring points on the evacuation path. When the path cannot be adjusted again, a dynamic planning scheme for the evacuation path of people in this fire situation is obtained. The scheme takes into account the architectural structure of the subway station and the changes in the fire environment and specifies the optimal evacuation paths for people at each moment.

CRediT authorship contribution statement

Zhaohui Liu: Conceptualization, Project administration, Resources, Supervision, Writing – review & editing. **Ruihong Zou:** Conceptualization, Formal analysis, Methodology, Software, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

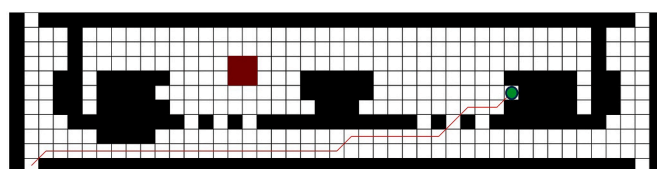


Fig. 19. Adjustment results of evacuation path on the station hall floor after 112s of fire.

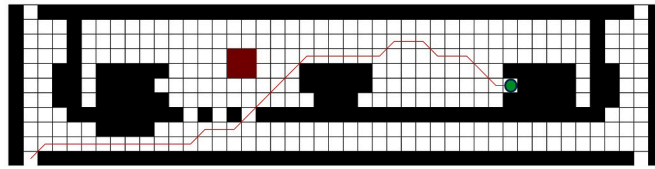
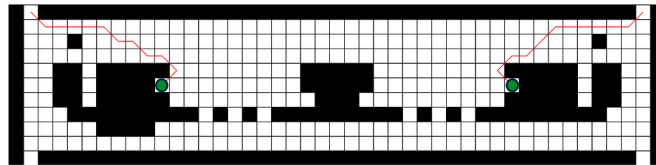
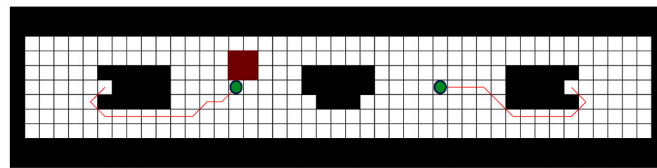


Fig. 20. Adjustment results of evacuation path on the station hall floor after 128s of fire.



(a) Station hall floor evacuation paths



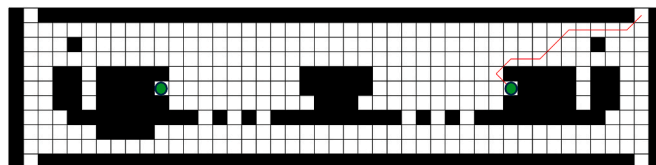
(b) Platform floor evacuation paths

Fig. 21. Optimal evacuation path for fire personnel on the platform floor.

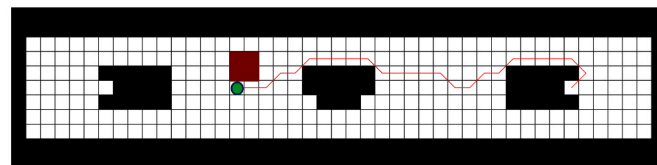
Table 9

The optimal evacuation path length for station platform fires.

Path	Evacuation path	Distance(m)
③	A—Staircase 1—C—Exit 1	51.456
④	B—Staircase 2—D—Exit 2	51.456



(a) Station hall floor evacuation paths



(b) Platform floor evacuation paths

Fig. 22. Evacuation path adjustment results of station platform floor after 100s of fire.

Nomenclature

q	heat release rate of the ignition source (KW)
a	fire development coefficient (KW/s^2)
t	time(s)
$f_1(T)$	influence coefficient of temperature
$f_2(\rho)$	influence coefficient of carbon monoxide

$f_3(K_c)$	influence coefficient of visibility
T_0	room temperature (°C)
T_s	temperature at fire scene (°C)
T_{c1}	temperature causing discomfort (°C)
T_{c2}	temperature causing an injury (°C)
T_d	lethal temperature (°C)
v_0	normal walking speed (m/s)
v_{max}	maximum walking speed (m/s)
ρ_{CO}	volumetric concentration of CO (%)
K_c	light reduction coefficient
α	information heuristic factor
β	expected heuristic factor
ρ	pheromone volatilization factor
Q	pheromone constant
m	number of ants
$\eta_{ij}(t)$	degree of expectation of ants from node i to node j
d_{ij}	geometric distance between nodes i to node j
$P_{ij}(t)$	probability of selecting the node
$\tau_{ij}(t)$	pheromone concentration on path (i, j)
$allowed_k$	nodes set
s	a node belonging to nodes set
L_k	total length of the path traveled ant k
K	number of iterations
K_{Max}	maximum number of iterations

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